

Experimental and Theoretical Study of the $\text{C}_2\text{H}_3 \rightleftharpoons \text{H} + \text{C}_2\text{H}_2$ Reaction. Tunneling and the Shape of Falloff Curves

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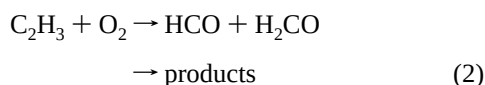
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The kinetics of the unimolecular decomposition of the C_2H_3 radical has been studied. The reaction was isolated for quantitative study in a heated tubular flow reactor coupled to a photoionization mass spectrometer. Rate constants for the decomposition were determined in time-resolved experiments as a function of temperature (879–1058 K) and bath gas density ($(6\text{--}48) \times 10^{16}$ molecules cm^{-3}) in He, Ar, and N_2 . The rate constants are close to the low-pressure limit under the conditions of the experiments. The potential energy surface and properties of the transition state were studied by ab initio methods. Experimental results of the current and earlier studies of both the direct and reverse reactions were analyzed and used to create a transition state model of the reaction. Falloff behavior was reproduced using master equation modeling with parameters obtained from optimization of the agreement between experimental and calculated rate constants. The effects of tunneling on the shape of falloff and the values of the low-pressure-limit rate constants were investigated. It was demonstrated that these effects exceed those for the high-pressure-limit rate constants by orders of magnitude and cannot be neglected. The resulting model of the reaction provides the high-pressure-limit rate constants for the decomposition reaction ($k_1^\infty(\text{C}_2\text{H}_3 \rightarrow \text{H} + \text{C}_2\text{H}_2) = 3.86 \times 10^8 T^{1.62} \exp(-18650 \text{ K}/T) \text{ s}^{-1}$) and the reverse reaction ($k_{-1}^\infty(\text{H} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_3) = 6.04 \times 10^{-14} T^{1.09} \exp(-1328 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$). Values of $\langle \Delta E \rangle_{\text{all}} = -74$ (He), -115 (Ar), and -117 cm^{-1} (N_2) for the average energy loss per collision were obtained using an exponential-down model. Parametrization of the temperature and pressure dependence of the unimolecular rate constant for the temperature range 200–2120 K and pressures 1– 10^4 Torr in He and N_2 is provided using the modified Lindemann–Hinshelwood expression.

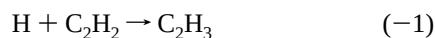
I. Introduction

Vinyl radicals are recognized as important intermediates in hydrocarbon combustion processes with elementary reactions of C_2H_3 influencing both the rate and products of the overall combustion process. Two major reactive pathways of vinyl radicals at high temperatures are



which lead to different fates for the carbon contained in vinyl radicals under combustion conditions.

Over the past three decades a number of experimental and theoretical studies have investigated the kinetics and mechanism of reaction 1 and the reverse reaction



(see refs 1–10 and references therein). No direct experimental determinations of the rate constants of reaction 1 are available in the literature. Among the indirect studies, the most rigorous is that of Rao and Skinner,¹⁰ who applied Lyman- α absorption to monitor H and D atoms behind shock waves. Error analysis performed by these authors indicated that the rate constants of reactions 1 and -1 were measured only with order-of-magnitude accuracy because of the unknown rates of other reactions involved.

Several groups experimentally studied the addition of H atoms to acetylene under conditions where the initial concentrations

of hydrogen atoms were low enough to ensure the lack of influence of fast secondary reactions on the measured primary reaction rate.^{2–7} Temperatures employed in these experiments were in the region 193–473 K, and pressures ranged from 1 to 1200 Torr.

Two major attempts^{3,8} to reproduce the experimental temperature and pressure dependencies of the rate constants of reaction -1 resulted in calculated rate constant values that were at least an order of magnitude higher than experimental ones. Both Keil et al.³ and Harding et al.⁸ had to introduce transmission coefficients ($\ll 1$) to achieve agreement between theory and experiment.

Tsang and Hampson¹¹ provided recommendations for the low-pressure-limit rate constants and falloff behavior of reaction 1 based on the RRKM analysis of the reverse reaction and the thermochemistry of vinyl radical. Due to the large uncertainty in the heat of formation of C_2H_3 , these authors estimated an uncertainty factor of 10 for the recommended rate constant values.

In the current study we present the first direct experimental determination of the rate constants of reaction 1. Rate constants were measured in three bath gases (He, Ar, and N_2) over the density range of $(6\text{--}48) \times 10^{16}$ molecules cm^{-3} and in the temperature range 879–1058 K. The experimental method employed has been successfully used before for measuring rates of unimolecular decomposition of radicals (ref 12 and references therein). Data analysis involved creation of a transition state model and master equation modeling of rate constants. Significant effects of tunneling on the shape of falloff curves and values of the low-pressure rate constants were investigated. Parameters of the model were obtained from our ab initio study and from fitting of our experimental results and those available in the literature. The results of modeling provide recommenda-

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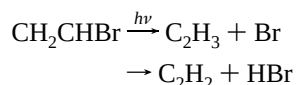
tions for the high-pressure-limit rate constants of reactions 1 and -1 as well as information on weak collision effects.

II. Experimental Study

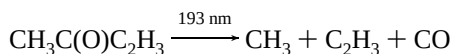
Details of the experimental apparatus¹³ and procedures¹⁴ used have been described before and so are only briefly reviewed here. The vinyl radicals were produced homogeneously at elevated temperatures in a heatable quartz tubular reactor by pulsed laser photolysis and their unimolecular decay subsequently monitored in time-resolved experiments using photoionization mass spectrometry.

Pulsed unfocused 193 or 248 nm radiation (≈ 5 Hz) from a Lambda Physik EMG 201MSC excimer laser was directed along the axis of a heatable quartz reactor (1.05 cm i.d.) coated with boron oxide. Gas flowing through the tube at ≈ 4 m s⁻¹ contained the radical precursor ($<0.01\%$) and the inert carrier gas (He, Ar, or N₂) in large excess. The flowing gas was completely replaced between laser pulses.

Vinyl radicals were produced by the pulsed, 193- or 248-nm laser photolysis of vinyl bromide¹⁵



or 193-nm photolysis of methyl vinyl ketone¹⁶



Initial conditions (precursor concentration and laser intensity) were selected to provide low radical concentrations ($\leq 10^{11}$ molecules cm⁻³). Under these conditions reactions between radical products (including $\text{C}_2\text{H}_3 + \text{C}_2\text{H}_3$) had negligible rates compared to that of the unimolecular decomposition of the C_2H_3 radicals.

Gas was sampled continuously through a conical hole (0.043-cm diameter at the inside of the reactor) in the side of the reactor and formed into a beam by a skimmer before the gas entered the vacuum chamber containing the photoionization mass spectrometer. As the gas beam traversed the ion source, a portion was photoionized and mass selected. Vinyl radicals were ionized using the light from a hydrogen resonance lamp (10.2 eV) with a MgF₂ window. Temporal ion signal profiles were recorded on a multichannel scaler from a short time before each laser pulse up to 20 ms following the pulse. Data from 1000 to 28000 repetitions of the experiment were accumulated before the data were analyzed.

The gases used were obtained from Aldrich (methyl vinyl ketone, 99%) and Matheson (He, >99.995%; Ar, >99.998%; N₂, >99.998%; vinyl bromide, 99.5%). Radical precursors were purified by vacuum distillation prior to use. Helium, argon, and nitrogen were used as provided.

The vinyl ion signal profiles were fit to an exponential function ($[\text{C}_2\text{H}_3]_t = [\text{C}_2\text{H}_3]_0 e^{-k't}$) using a nonlinear least-squares procedure. Experiments were performed to establish that the decay constants did not depend on the nature or concentration of the precursor (provided that the concentration was kept low enough to ensure that radical-radical reactions had negligible rates), the laser intensity, or wavelength. The exponential decay constants depended only on temperature and bath gas density.

The unimolecular decay of C_2H_3 was observed only above 700 K. Below this temperature a slow decay of C_2H_3 is observed due to a first-order heterogeneous wall-loss process:

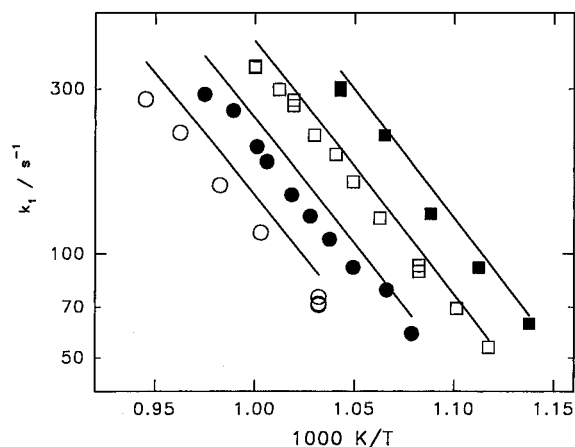
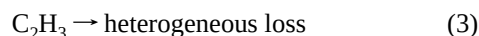


Figure 1. Plot of C_2H_3 unimolecular rate constants (k_1 vs $1000/T$) for different He densities (atoms cm⁻³). Lines represent values obtained from recommendation of Tsang and Hampson¹¹ assuming the same collision efficiency for He as for N₂ as bath gas: open circles, $[\text{He}] = 6 \times 10^{16}$ atom cm⁻³; closed circles, $[\text{He}] = 12 \times 10^{16}$ atom cm⁻³; open squares, $[\text{He}] = 24 \times 10^{16}$ atom cm⁻³; closed squares, $[\text{He}] = 48 \times 10^{16}$ atom cm⁻³.

Above 700 K the decay constant increased rapidly with rising temperature due to the increasing importance of the thermal decomposition of the vinyl radical, reaction 1. The C_2H_3 decay constants were analyzed assuming that the vinyl radicals were consumed only by two elementary reactions, (1) and (3). At low temperatures, only the heterogeneous loss is observed ($k' = k_3$). Above 700 K the sum of the two loss processes is observed ($k' = k_1 + k_3$).

Calculations of k_1 from measurements of k' require knowledge of k_3 above 700 K. While k_3 was directly determined below 700 K ($18\text{--}47$ s⁻¹, independent of temperature), it could not be measured above this temperature due to the additional loss of radicals by unimolecular decomposition. Values of k_3 above 700 K needed to determine k_1 from the measurements of k' were obtained by an extrapolation assuming that k_3 retains its temperature independence beyond 700 K up to the highest temperature of this study, 1058 K. To minimize possible errors in the determination of k_1 caused by this assumed temperature independence of k_3 above 700 K, experiments to obtain k_1 were conducted at temperatures sufficiently high to assure that $k' \geq 3k_3$. It was this criterion that established the lowest temperature used to determine k_1 at each bath gas density. The highest temperature used at each total gas density was determined by the fact that decay constants above 400 s⁻¹ could not be measured accurately.

The results of all the experiments and conditions used to determine k_1 are given in Table 1. The unimolecular rate constants for reaction 1 obtained from these sets of experiments conducted at different densities are shown in Figures 1 and 2. Estimated uncertainties in the k_1 determinations vary from $\pm 10\%$ in the middle of the temperature range to $\pm 20\%$ at the extreme temperatures used. Also in Figure 1 are shown rate constants which were calculated using formulas recommended by Tsang and Hampson¹¹ taking the same value for collision efficiency for He as that recommended for N₂ by these authors. As can be seen from the plot, there is a good agreement between experimental and predicted data.

III. Tunneling and the Shape of Falloff Curves

Although tunneling is frequently taken into account in theoretical treatments of bimolecular reactions (see, for example, refs 17 and 18 and references therein), it is generally ignored

TABLE 1: Conditions and Results of Experiments To Measure the Unimolecular Rate Constants (k_1) of the Thermal Decomposition of Vinyl Radicals in He, Ar, and N_2 as Bath Gases

$10^{-16}[M]$ (molecules cm^{-3})	T (K)	$10^{-11}[\text{precursor}]$ (molecules cm^{-3})	k_3 (s^{-1})	k_1 (s^{-1})
He as Bath Gas				
6	969	5.7	30.3	71.6
6	969	2.5	20.4	74.8
6	969	2.5	37.2	70.8
6	997	4.5	24.3	115.3
6	1018	4.4	25.7	158.3
6	1039	4.5	24.3	224.0
6	1058	4.5	24.3	280.1
12	927	3.1 ^a	22.8	58.7
12	938	4.9	21.7	78.5
12	953	3.1 ^a	22.8	91.4
12	964	4.9	21.7	110.4
12	973	3.1 ^a	22.8	128.7
12	982	4.9	21.7	148.6
12	994	3.1 ^a	22.8	185.0
12	999	4.9	21.7	204.0
12	1011	4.1	25.8	260.1
12	1026	3.1 ^a	22.8	289.5
24	895	56.7 ^b	18.1	53.6
24	908	4.0	25.8	69.3
24	924	4.0	25.8	88.9
24	924	56.7 ^b	18.1	92.4
24	941	4.0	25.8	127.0
24	953	56.7 ^b	18.1	161.8
24	961	4.0	25.8	193.7
24	971	56.7 ^b	18.1	220.2
24	981	4.0	25.8	278.4
24	981	4.0	25.8	268.7
24	988	56.7 ^b	18.1	298.8
24	1000	4.0	25.8	346.2
24	1000	4.0	40.7	349.0
48	879	4.1	25.1	62.6
48	899	4.1	25.1	91.2
48	919	4.1	25.1	130.7
48	939	4.1	25.1	219.9
48	959	4.1	25.1	296.8
48	959	4.1	39.1	304.0
Ar as Bath Gas				
12	952	5.8	46.9	97.4
12	971	5.8	46.9	118.5
12	991	5.8	46.9	188.4
N_2 as Bath Gas				
12	944	5.7	44.5	90.6
12	965	5.7	44.5	145.1
12	984	5.7	44.5	200.7
24	895	4.3	27.9	69.0
24	928	4.3	27.9	120.2
24	953	4.3	27.9	195.1

^a Methyl vinyl ketone was used as radical precursor. Vinyl bromide was used in all other experiments. ^b The 248-nm photolysis was used; 193-nm photolysis was used in all other experiments.

in modeling unimolecular rate constants in the falloff region. Miller¹⁹ and Kato and Morokuma²⁰ have developed a simple method of including tunneling in the RRKM method of calculation of energy-dependent rate constants $k(E)$. Although this approach has been applied in a strong-collision modeling of reaction -1 at room temperature,⁵ the effects of tunneling on the shapes of falloff curves and their temperature dependence have not been studied. We show here that these effects are significant and cannot be neglected.

Miller¹⁹ and Kato and Morokuma²⁰ showed that, if separability of motion along the reaction coordinate from other degrees of freedom of a reacting molecule is assumed, tunneling can be accounted for by a modification of the RRKM formula for the energy-dependent rate constant $k(E)$:

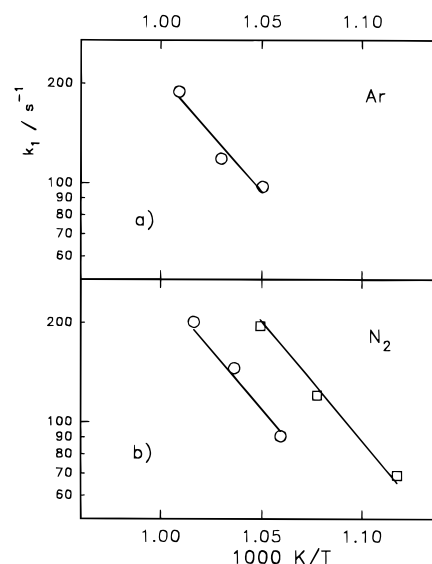


Figure 2. Plot of C_2H_3 unimolecular rate constants (k_1 vs $1000/T$) for Ar and N_2 as bath gases. Lines represent the results of master equation simulations using our best model (see section IV) with $\langle \Delta E \rangle_{all} = -115$ cm^{-1} for Ar and -117 cm^{-1} for N_2 . Open circles, $[M] = 6 \times 10^{16}$ molecule cm^{-3} ; open squares, $[N_2] = 12 \times 10^{16}$ molecule cm^{-3} .

$$k(E) = \frac{W^\ddagger(E)}{h\rho(E + E_1)}$$

Here $W^\ddagger(E)$ is the sum of states of the transition state, $\rho(E + E_1)$ is the density of states of the reacting molecule, and E is energy relative to the reaction barrier E_1 . In this modification, $W^\ddagger(E)$ is substituted with $W_Q^\ddagger(E)$ obtained by a convolution of $W^\ddagger(E)$ and $P'(E)$, the first derivative of the energy-dependent transition probability $P(E)$:

$$W_Q^\ddagger = \int_{-E_1}^E P'(\epsilon) W^\ddagger(E - \epsilon) d\epsilon$$

Mathematically, this is equivalent to adding a new “tunneling degree of freedom” to the transition state. $P(E)$ serves as a sum-of-states function of this “degree of freedom” and $P'(E)$ as a density-of-states function. The high-pressure limit tunneling correction factor¹⁹ $\kappa(T)$

$$\kappa(T) = \int_{-E_1}^{\infty} P'(E) e^{-E/k_B T} dE \quad (I)$$

is equivalent to the partition function of this “tunneling degree of freedom” in the transition state theory formula for the rate constant.

We illustrate the influence of tunneling on the shape of falloff curves using as an example reactions 1 and -1 , the unimolecular decomposition of vinyl radical and the reverse reaction of H atom addition to acetylene. For this purpose we use our model of reactions 1 and -1 which will be described later (section IV).

The influence of tunneling on the shape of the $k(E)$ function was shown by Miller for two reactions,



and



as examples. At energies above the barrier $k(E)$ dependencies are quite similar in the absence and in the presence of tunneling. Below the barrier, in the absence of tunneling one can see a

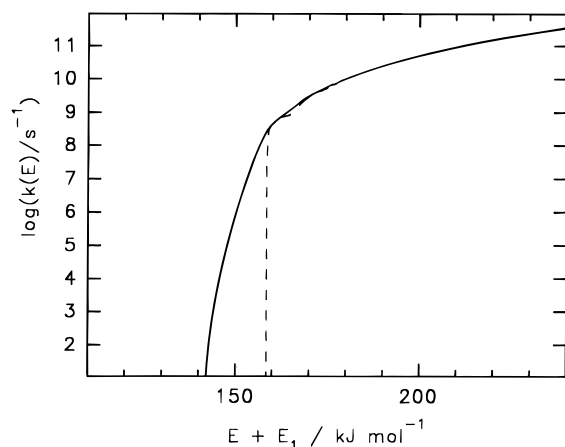


Figure 3. Energy-dependent rate constants of reaction 1 ($k(E)$ vs energy). Solid line, with tunneling (section III); dashed line, without tunneling.

sharp cutoff of $k(E)$ at an energy equal to the barrier height. If tunneling is taken into account, this cutoff is replaced by a slower decline of $k(E)$ with decreasing energy. This decline, however, is considerably steeper than $k(E)$ dependence above the barrier. An analogous dependence can be seen in the case of reaction 1 (Figure 3). There is, however, a difference between reactions 1 and -1 and reactions 4a and 4b. Both reactions considered by Miller had barriers for the reverse reactions higher than or comparable to the barrier for the direct reaction. In our model of reaction 1, the barrier for the reverse reaction ($E_{-1} = 16.9 \text{ kJ mol}^{-1}$) is smaller than that for the direct reaction ($E_1 = 158.6 \text{ kJ mol}^{-1}$). Thus, $k(E)$ for reaction 1 exhibits a cutoff, not at $E = E_1$, but at $E = \Delta E_{1,-1} = E_1 - E_{-1}$ with a steady steep decline between these two values of energy (Figure 3).

One effect of this change in the $k(E)$ function on the shape of falloff curves of unimolecular reactions will be a widening of falloff because of its extension toward lower pressures. In the absence of tunneling, at some pressure corresponding to the low-pressure limit the collisional energy transfer rate is much lower than $k(E)$ values above the barrier. Thus, collisional energy transfer cannot compete with the reaction, the population of molecules above the barrier is equal to zero, and the shape of the population below the barrier is independent of pressure.²¹ In the presence of tunneling, at the same pressure the collisional energy transfer still may be comparable to a lower value of $k(E)$ at some energy below the barrier. In this case the shape of the population below the barrier will depend on pressure and the reaction will not be in the low-pressure limit.

Another consequence of tunneling is that the values of the low-pressure-limit rate constant will be determined by collisional energy transfer,²¹ not to the reaction barrier E_1 , but to the lower $\Delta E_{1,-1} = E_1 - E_{-1}$. Hence, one can expect significantly higher values of k^0 .

The effects of the extension of falloff toward lower pressures are illustrated in Figure 4, which shows the dependencies of k_1 vs $[\text{He}]$ calculated with (solid lines) and without (dashed lines) accounting for tunneling. It can be seen that while in both cases the high-pressure limit is reached at the same bath gas density, the low-pressure limit in the case of tunneling is reached at significantly lower pressures. The same data are shown in Figure 5 in the form of reduced falloff curves (k_1/k_1^∞ vs $k_1^0[\text{M}]/k_1^\infty$) which illustrates a notable widening of falloff.

There is a significant difference between the effects of tunneling on the high-pressure-limit and low-pressure-limit rate constants. The latter can be orders of magnitude greater than

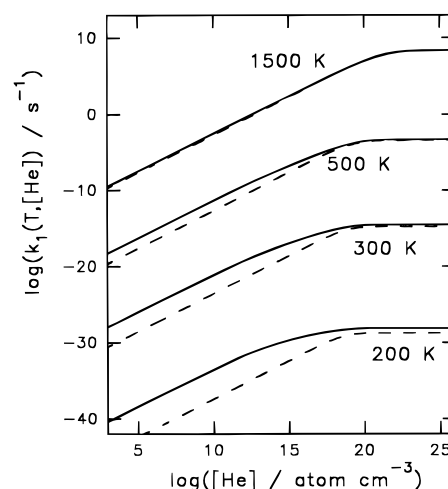


Figure 4. Falloff curves for reaction 1 ($\log(k_1)$ vs $\log([\text{He}])$) at different temperatures calculated with (solid lines) and without (dashed lines) accounting for tunneling.

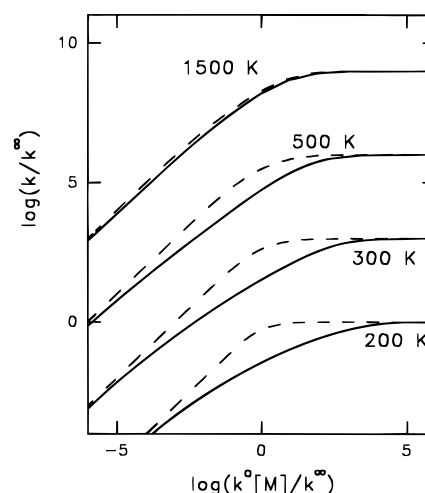


Figure 5. Same as in Figure 4 in reduced coordinates ($\log(k_1/k_1^\infty)$ vs $\log(k_1^0[\text{He}]/k_1^\infty)$). Curves calculated for different temperatures are shifted upward by successive factors of 1000.

the former. This is illustrated in Figure 6, which shows $k_{-1}^\infty(T)$ and $k_{-1}^0(T)$ dependencies obtained for reaction -1 using the model described in section IV. The values of k_{-1} in all portions of falloff were calculated from the corresponding values of k_1 and the equilibrium constant.^{21,22} One can see that low-pressure effects of tunneling significantly exceed the high-pressure ones. For example, at 200 K the difference between the high-pressure rate constants calculated with and without accounting for tunneling is only a factor of 4.2, while the same factor for the low-pressure constant is 1.0×10^4 . As one goes to higher temperatures and the width of the energy distribution of reacting molecules increases, the "perturbation" of the $k(E)$ function caused by tunneling becomes less important and the effects of tunneling necessarily decrease. The low-pressure effects, however, still remain significant at higher temperatures. Even at 1000 K, where the high-pressure tunneling effect is hardly noticeable (only 6%), the difference between the low-pressure-limit rate constants is still a factor of 3.1.

The above example demonstrates the necessity of properly accounting for tunneling in the modeling of falloff in reactions involving the transfer of a light atom, even at temperatures high enough to make the contribution of tunneling at the high-pressure-limit negligible.

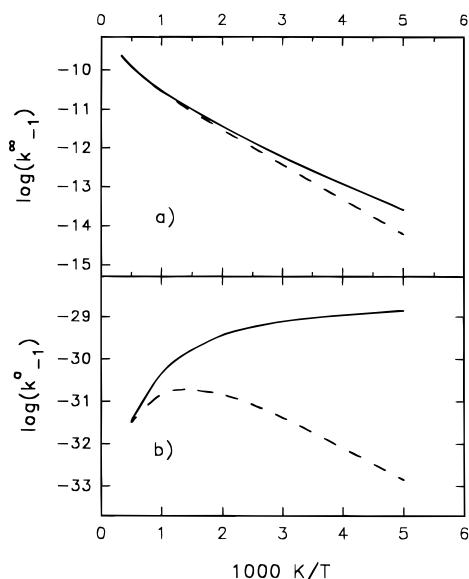


Figure 6. Temperature dependencies of k_{-1}^{∞} ($\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$) and k_{-1}^0 ($\text{cm}^6 \text{ molecule}^{-2} \text{ s}^{-1}$) with (solid lines) and without (dashed lines) accounting for tunneling.

IV. Data Analysis

In this section we present the development of a model of reactions 1 and -1 which describes both our high-temperature, low-pressure data on reaction 1 and the body of low-temperature data on reaction -1. Data available in the literature are first reviewed and analyzed. Properties of the C_2H_3 decomposition transition state are determined from an ab initio study. These ab initio results are used in a model of reactions 1 and -1, the parameters of which (such as energy barriers and temperature dependencies of the collisional energy transfer parameters) are adjusted to reproduce the bulk of available experimental data. The effects of tunneling, which have been shown to be essential (section III), are included in the calculation of energy-dependent rate constants $k(E)$ by the RRKM method. The falloff behavior is analyzed using the master equation. Finally, the thermochemistry of C_2H_3 and the uncertainties of the model parameters are evaluated.

IV.1. Literature Data on the Rate Constants of Reactions 1 and -1. Most of the available data have been previously reviewed in the literature.^{3,8,23} No direct experimental determinations of the rate constants of reaction 1 are available. Benson and Haugen²⁴ analyzed the data of Skinner and Sokoloski²⁵ on the hydrogenation of acetylene to give a recommendation for the low-pressure-limit rate constant of reaction 1. However, according to Rao and Skinner,¹⁰ these data were not highly sensitive to the rate constants of C_2H_3 decomposition. Among the indirect studies, the most rigorous is that of Rao and Skinner,¹⁰ who applied Lyman- α absorption to monitor H and D atoms behind incident and reflected shock waves. These authors studied reaction 1 at temperatures 1060–1750 K and pressures of 0.3–0.5 atm using the pyrolysis of vinyl bromide or vinyl iodide as a source of vinyl radicals. They also studied the formation of D atoms in the reaction of H atoms with C_2D_2 at temperatures of 1200–1900 K and pressures of 0.4–3.2 atm using the pyrolysis of neopentane or toluene as a source of H atoms. The values of the rate constants of the $\text{H} + \text{C}_2\text{D}_2 \rightarrow \text{D} + \text{C}_2\text{HD}$ reaction obtained in these studies were converted into k_{-1}^{∞} by assuming that stabilization of the excited CDCDH adduct is negligible and that the rate of its dissociation to form H atom is 1.8 times that to form D atom. Error analysis performed by Rao and Skinner indicated that rate constants of reactions 1 and -1 were measured only with the order-of-

magnitude accuracy because of the unknown rates of other reactions involved. Unfortunately, such high uncertainties practically preclude any potential use of these data in developing a model of reactions 1 and -1.

Although numerous experimental determinations of k_{-1} have been reported in the past, many of these data were obtained under experimental conditions where fast secondary reactions significantly influenced the measured values of k_{-1} . Only four groups experimentally studied reaction -1 using low enough initial concentrations of hydrogen atoms to avoid a contribution from secondary reactions.^{2–7} Payne and Stief² used flash photolysis to produce and resonance fluorescence to detect H atoms at $T = 193$ –400 K and pressures of 10–700 Torr of He. Keil et al.³ used the discharge flow method with mass spectrometry and time-resolved Lyman- α photometry at room temperature and pressures of 1–742 Torr. In this work, only data obtained with Lyman- α photometry were not influenced by secondary reactions, and data obtained in the mass spectroscopic experiments required a correction by a stoichiometric factor which was estimated by authors to be close to 2. Sato et al.^{4–6} used electron pulsed radiolysis with detection of H atoms by Lyman- α resonance absorption at temperatures of 207–451 K and pressures of 100–1300 Torr. Finally, Ellul et al.⁷ applied a pulsed Hg-sensitized photolysis–resonance absorption technique at a pressure of 1 atm and temperatures of 298–473 K. There is significant disagreement between the data of Sato et al. and those of all other studies. The results of Payne and Stief, Keil et al. (at room temperature), and Ellul et al. are in agreement with each other. Rate constants reported by Sato et al., however, are higher than those of these studies by approximately a factor of 2 at low temperatures and a factor of 3 at high temperatures. This disagreement is difficult to explain, especially considering that, using these same techniques, Sato et al. and Payne and Stief agreed on the high-pressure rate constants of the addition of H atoms to ethylene.^{6,26} The only possible explanation offered⁵ for this discrepancy is that the pulsed radiolysis method used by Sato et al. might produce inhomogeneous initial concentrations of H atoms that could be high enough to yield a stoichiometric factor of 2 (due to the fast secondary reaction $\text{H} + \text{C}_2\text{H}_3 \rightarrow \text{products}$).

Gordon et al.^{27,28} applied two H-maser techniques to study the high-pressure limit rate constant of reaction -1 at room temperature. These techniques, applied to the reversible addition of H atom to a molecule, enable measurement of the rate of the adduct formation (without the negative contribution of the reverse dissociation), i.e., determination of the high-pressure-limit rate constant. The value obtained, $k_{-1}^{\infty}(305 \text{ K}) = (3.5 \pm 0.2) \times 10^{-13} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$, is approximately twice as high as that reported by Payne and Stief.² It was suggested by Gordon et al. that this discrepancy may be accounted for by incomplete stabilization of the C_2H_3 formed in reaction -1 at the highest pressures used in ref 2 (i.e., the high-pressure limit was not reached). As will be shown later in this section, this suggestion is valid, and better agreement is achieved when falloff effects are taken into account.

The experimental results of Payne and Stief constitute the most accurate and comprehensive set of pressure- and temperature-dependent data on the rates of reaction -1. We therefore focus our modeling attempts on reproducing these data along with those obtained in our measurements of the rate constants of reaction 1.

IV.2. Ab Initio Study of the Transition State and the Shape of the Potential Energy Profile along the Reaction Path. The physical properties of C_2H_2 are known.^{29,30} For the vinyl radical, we use a set of frequencies and rotational constants

TABLE 2: Geometrical Structure^a and Energies of C₂H₃, C₂H₃[‡], and C₂H₂ Obtained in the ab Initio Study

property	C ₂ H ₃	C ₂ H ₃ [‡]	C ₂ H ₂
Structures (Å, deg)			
C1C2	1.288	1.200	1.219
H11C1	1.086	1.712	1.063
H12C1	1.082	1.064	
H21C2	1.077	1.062	1.063
H11C1C2	121.63	110.10	180.00
H12C1C2	122.05	159.83	
H21C2C1	136.41	171.56	180.00
Energies (hartrees)			
E(HF/6-31G**)	-77.394 251	-77.309 355	-76.819 456
E(MP2/6-31G**)	-77.627 982	-77.552 230	-77.081 668
E(MP4/6-31G**)	-77.663 531	-77.586 527	-77.109 140
E(PMP4/6-31G**)	-77.671 223	-77.597 181	

^a C₂H₃ and C₂H₃[‡] have planar structure; C₂H₂ is linear.

reported by Ervin et al.³¹ which is a combination of experimental measurements and ab initio calculations.

Harding et al.⁸ performed extensive high-level ab initio calculations of the structure, vibrational frequencies, and energies of C₂H₃, C₂H₂, and C₂H₃ decomposition transition state. The results of these authors on the geometry and vibrational frequencies of acetylene show excellent agreement with experiment. The information on the C₂H₃[‡] properties provided by Harding et al. is sufficient for calculating the energy-dependent rate constants $k(E)$ by the RRKM method. However, the treatment of tunneling applied in the current study (sections III and IV.3) requires knowledge of the effective width of the potential energy barrier of the reaction. While this barrier width (a geometrical parameter) can be obtained from the imaginary frequency calculated at the transition state, direct determination of this parameter from the potential energy profile along the reaction path is preferred. The imaginary frequency is determined by the width and the height of the reaction barrier. Ab initio methods usually overestimate the barrier height and, therefore, overestimate the curvature of the potential energy surface in the direction of the reaction coordinate and the imaginary frequency associated with the barrier. On the other hand, geometrical parameters are usually determined more accurately by the same methods. This makes the current method of the treatment of tunneling more accurate compared to the one where the imaginary frequency of the transition state obtained from ab initio calculations is used directly in determining the transition probability. The current ab initio study, therefore, focuses on the characterization of the shape of the potential energy barrier.

The geometries and harmonic vibrational frequencies of the vinyl radical decomposition transition state (C₂H₃[‡]) were studied using the ab initio UMP2 method with the 6-31G** basis. Energies were calculated with the MP4/6-31G** method. For the purpose of obtaining complete information on the shape of the potential energy surface along the reaction coordinate, properties of vinyl radical and acetylene were also calculated at different levels of theory. Structures and energies of these species are listed in Table 2. The GAUSSIAN 92 system of programs³² was used in all ab initio calculations.

The shape of the potential barrier for reactions 1 and -1 was calculated using the method of reaction path following (intrinsic reaction coordinate)³³ in mass-weighted internal coordinates described by Gonzalez and Schlegel (ref 33b). For each point along the reaction path, optimization was done at the UMP2/6-31G** level, and energy was calculated at the UMP4/6-31G** level. Spin contamination was removed by the spin projection (PMP4) method of Schlegel.^{34,35} The reaction coordinate

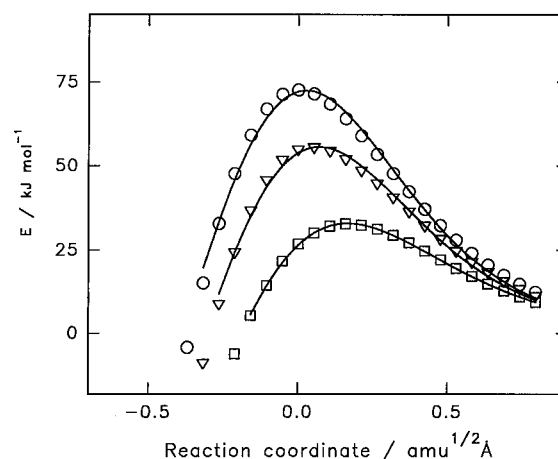


Figure 7. Shapes of potential energy barrier of reactions 1 and -1 obtained at different levels of calculation. Circles, UMP2; triangles, UMP4; squares, PMP4. Lines represent fits to Eckart function. Energy relative to H + C₂H₂.

TABLE 3: Eckart Parameters Obtained from Fitting the Potential Energy Profile along the Reaction Path

	UMP2	UMP4	PMP4
$A^a/\text{kJ mol}^{-1}$	126.24	147.44	167.64
$B/\text{kJ mol}^{-1}$	511.45	471.33	396.25
$l/\text{amu}^{1/2} \text{ Å}$	1.60	1.66	1.83
ν_i^b/cm^{-1}	1562	1388	1051
$E_{-1}^b/\text{kJ mol}^{-1}$	72.53	55.64	32.97

^a A was fixed at the value obtained from ab initio energies of C₂H₃, C₂H₂, and H. ^b Imaginary frequency ν_i and potential barrier for reaction -1 calculated from the fitted values of B and l .

direction nearly coincided with that of the C-H bond of the departing H atom.

The results of this ab initio study of the decomposition transition state C₂H₃[‡] and the reaction path were used in creating a model of reactions 1 and -1. Although the reaction energy threshold values calculated at the applied levels of theory are not believed to be sufficiently accurate, the geometrical configuration of the transition state (Table 2) was used in the model. The calculated set of vibrational frequencies of the transition state (scaled by 0.9) was used as a basis for the model.

The shapes of the potential energy surface obtained at different levels of calculations (UMP2, UMP4, and PMP4) were used to determine the width of the potential energy barrier—an important parameter required for the modeling of tunneling. The potential energy profile along the reaction path was fitted with the Eckart function

$$V = \frac{A\xi}{1 + \xi} + \frac{B\xi}{(1 + \xi)^2}; \quad \xi = \exp\left(\frac{2\pi x}{l}\right) \quad (\text{II})$$

where x is a coordinate along the reaction path, l is a parameter determining the width of the barrier, and parameters A and B are related to the barriers for the direct and reverse reactions E_1 and E_{-1} :

$$A = \Delta E_{1,-1} = E_1 - E_{-1}; \quad B = (\sqrt{E_1} + \sqrt{E_{-1}})^2 \quad (\text{III})$$

The transition probability for such a barrier can be described analytically, as shown by Eckart.³⁶ In the fitting process, parameter A was fixed at the value obtained from ab initio calculations, and B and l were determined from the fitting. Only points with energy above that of H + C₂H₂ were used (Figure 7). The resultant values of B , l , and E_{-1} are listed in Table 3 for three levels of calculations used. Although it can clearly

be seen that improvement of the level of theory results in a significant reduction of the reverse reaction barrier, the barrier width changes only slightly—from 1.60 amu^{1/2} Å at the UMP2 level to 1.66 amu^{1/2} Å at the UMP4 level, and to 1.83 amu^{1/2} Å at the PMP4 level. In the model of reactions 1 and -1 we use the PMP4 value, and the change of model parameters with the variation of l by 0.23 amu^{1/2} Å (the difference between values obtained at PMP4 and UMP2 levels) is investigated.

An alternative method is to calculate the value of the Eckart barrier width parameter l from the values of the imaginary frequency calculated at the transition state and A and E_{-1} parameters obtained in the same ab initio calculations. For reactions 1 and -1 this approach gives results close to the above values of l obtained in the fitting of the potential energy profile. The UMP2 level unscaled imaginary frequency $\nu_i = 1580$ cm⁻¹ combined with the reaction energetics obtained at the same level yields $l = 1.59$ amu^{1/2} Å. A similar result can be obtained from the ab initio values of Harding et al.⁸ ($\nu_i = 1177$ cm⁻¹, $E_1 = 185.8$ kJ mol⁻¹, $E_{-1} = 31.8$ kJ mol⁻¹), which give $l = 1.60$ amu^{1/2} Å. This agreement between the values of l obtained from calculations at the transition state only and those derived from the reaction path analysis, however, is not universal. In a similar treatment of the $C_2H_3 + H_2 \rightarrow H + C_2H_4$ reaction,³⁷ the discrepancy between the l values obtained by these two different methods is significant: $l = 1.64$ amu^{1/2} Å from the reaction path analysis and $l = 1.26$ amu^{1/2} Å from the imaginary frequency.

The values of E_{-1} and ν_i obtained at the best level of ab initio calculations used here ($E_{-1} = 33.0$ kJ mol⁻¹ at the PMP4/6-31G** level, $\nu_i = 1051$ cm⁻¹ from the Eckart barrier fitting) are close to those reported by Harding et al.⁸ ($E_{-1} = 31.8$ kJ mol⁻¹, $\nu_i = 1177$ cm⁻¹). The agreement in the vibrational frequencies, however, is noticeably worse. The current study yields the values of 512, 701, and 745 cm⁻¹ for the three lowest frequencies which are significantly higher than 422, 470, and 678 cm⁻¹ reported by Harding et al.

IV.3. Transition State Model of Reactions 1 and -1 and Master Equation Modeling of Falloff. The rate constants of reactions 1 and -1 obtained in the current study and that of Payne and Stief² were reproduced using master equation simulations. The “exponential down” model³⁸ of collisional energy transfer was adopted to characterize the probability of deactivation of the C_2H_3 molecule with a total internal energy E in active degree of freedom to a final energy E' by collision with a molecule of bath gas:

$$P(E', E) = C \exp(-[E - E']/\langle \Delta E \rangle_{\text{down}}), \quad E' < E$$

$\langle \Delta E \rangle_{\text{down}}$ is the average energy lost in deactivating collisions, and C is a normalization constant.

It has been shown that accounting for the dependence of $\langle \Delta E \rangle_{\text{down}}$ on E results in a change of shape of the falloff curve.³⁹ We express this dependence in the form $\langle \Delta E \rangle_{\text{down}}(E) = \langle \Delta E \rangle_{\text{down}}^0(E/E_1)$ where E_1 is the barrier to dissociation and $\langle \Delta E \rangle_{\text{down}}^0$ is the value of $\langle \Delta E \rangle_{\text{down}}$ at $E = E_1$. This proportional functional form of the dependence of $\langle \Delta E \rangle_{\text{down}}$ on energy results in an approximate dependence of $\langle \Delta E \rangle_{\text{all}} \propto E^2$ ($\langle \Delta E \rangle_{\text{all}}$ is the average energy lost in both activating and deactivating collisions), which is appropriate for small molecules.^{38b,40} For the purpose of simplicity, the $\langle \Delta E \rangle_{\text{down}}$ and $\langle \Delta E \rangle_{\text{all}}$ values given later in this article represent those values at energy equal to the barrier height.

The energy-dependent rate constants $k(E)$ were calculated using the RRKM method.^{21,41} The density-of-states and sum-of-states functions of the vinyl radical and the transition state needed for the calculation of $k(E)$ values and further analysis

TABLE 4: Models of the Molecules and Transition State Used in the Data Analysis (Our Best Model, $\langle \Delta E \rangle_{\text{all}} = -74$ cm⁻¹ in He)

Energy Barriers					
$E_1 = 158.6 \text{ kJ mol}^{-1}$; $E_{-1} = 16.9 \text{ kJ mol}^{-1}$; $\Delta E_{1,-1} = 141.7 \text{ kJ mol}^{-1}$					
Vibrational Frequencies (cm ⁻¹)					
C ₂ H ₃ ^a	3265, 3190, 3115, 1670, 1445, 1185, 920, 825, 785				
C ₂ H ₃ ^{‡ b}	3265, 3151, 2081, 963, 835, 745, 701, 512, 771i				
C ₂ H ₂ ^c	3496, 3419, 1992, 764 (2), 625 (2)				
Rotational Constants (cm ⁻¹) and Symmetry Numbers					
C ₂ H ₃ ^a	7.49 (1, active); 1.07 (1, inactive); 0.93 (1, inactive)				
C ₂ H ₃ ^{‡ b}	6.45 (1, active); 1.09 (1, inactive); 0.93 (1, inactive)				
C ₂ H ₂ ^c	1.18 (2)				
Lennard-Jones Parameters ^d					
	He	Ar	N ₂	H ₂	C ₂ H ₃
σ (Å)	2.551	3.542	3.798	2.827	4.163 ^e
ϵ/k (K)	10.22	93.3	71.4	59.7	224.7 ^e

^a Properties of C_2H_3 from ref 31. ^b Structure and vibrational frequencies of $C_2H_3^\ddagger$ from our ab initio study. Imaginary frequency calculated from Eckart potential parameters of our best model. ^c Acetylene parameters from refs 29 and 30. ^d From ref 42. ^e Taken to be the same as for C_2H_4 .

are obtained from the models of these species described above. (Vinyl radical properties were taken from Ervin et al.,³¹ and those of the transition state were from our ab initio study; section IV.1, Table 4.) The density of states was calculated using the modified Beyer–Swinehart algorithm.^{21,43} The sum-of-states function for the transition state was similarly calculated. Tunneling was accounted for using the algorithm of Miller¹⁹ (see section III). The energy-dependent transition probability $P(E)$ was obtained from the formula of Eckart.³⁶ Since the use of intrinsic reaction coordinate (IRC) in mass-scaled coordinates reduces the problem of the motion along the reaction coordinate to that of the motion of a unit mass,³³ the effective reduced mass associated with tunneling is included in the parameter l (which has a dimension of amu^{1/2} Å).

The method of Gaynor et al.^{21,44} for solving the master equation was employed. The energy bin size used in the fitting procedures was 20–50 cm⁻¹. By comparing the exact values of the partition functions and high-pressure-limit rate constants with those obtained by numerical integration, it was found that errors resulting from numerical convolution of the sum-of-states function, as well as from numerical integration, are not more than 3% of the exact values at the temperatures of the experiments.

Barriers for the direct and reverse reaction (E_1 and E_{-1}), as well as energy transfer parameters ($\langle \Delta E \rangle_{\text{down}}$ and $\langle \Delta E \rangle_{\text{all}}$), were obtained in a fitting procedure. The goal of this fitting was to reproduce available information on the reaction rate data at both low and high temperatures. The results of Payne and Stief² (which constitute the most comprehensive set of pressure- and temperature-dependent data confirmed by other workers) were chosen as representative of the low-temperature, medium-pressure data. In particular, attention was focused on data obtained at the temperature of 228 K since they are the closest to the high-pressure limit. High-temperature rate data were represented by our experimental results at $T = 879$ – 1058 K and $[He] = (6$ – $48) \times 10^{16}$ atoms cm⁻³. Since k_1 vs $[He]$ dependence (deviation from the low-pressure limit) could not be reproduced exactly (vide infra), we focused our attention on reproducing the data obtained at $[He] = 24 \times 10^{16}$ atoms cm⁻³, which is in the middle of the experimental pressure range. These data are very close to the low-pressure limit, and hence, their modeling is most sensitive to the barrier for decomposition (E_1)

TABLE 5: Parameters Obtained from Fitting Experimental Data on k_1 and k_{-1} in He Using Different Temperature Dependencies of $\langle\Delta E\rangle_{\text{down}}$

temp dependence of $\langle\Delta E\rangle_{\text{down}}$	$\langle\Delta E\rangle_{\text{down}} = \text{constant}$		$\langle\Delta E\rangle_{\text{down}} = \alpha T$		$\langle\Delta E\rangle_{\text{all}} = \text{constant}$
	fit to k_{-1} data ^a	fit to k_1 data ^b	fit to k_{-1} data ^a	fit to k_1 data ^b	
$E_1/\text{kJ mol}^{-1}$	151.4	164.0	166.3	154.5	158.6
$E_{-1}/\text{kJ mol}^{-1}$	16.9	17.6	16.9	15.7	16.9
$\Delta E_{1,-1}/\text{kJ mol}^{-1}$	134.5	146.4	149.4	138.8	141.7
$\Delta H_{\text{f}}^{\circ}{}_{298}(\text{C}_2\text{H}_3)/\text{kJ mol}^{-1}$	306.1	294.2	291.2	301.8	298.9
$\langle\Delta E\rangle_{\text{down}}/\text{cm}^{-1}$	210	450	0.548T/K	0.260T/K	
$\langle\Delta E\rangle_{\text{all}}/\text{cm}^{-1}$					-74 ^c

^a Shape of falloff in k_{-1} at 228 K is reproduced while agreement with high-temperature k_1 data is reached only in the middle of temperature interval. ^b Temperature dependence of k_1 is reproduced while agreement with k_{-1} vs P data at 228 K is reached only in the middle of pressure interval. ^c Constant $\langle\Delta E\rangle_{\text{all}} = -74 \text{ cm}^{-1}$ corresponds to $\langle\Delta E\rangle_{\text{down}}$ increasing with temperature ($\langle\Delta E\rangle_{\text{down}} = 149, 220, 322, \text{ and } 640 \text{ cm}^{-1}$ at 200, 500, 1000, and 2000 K, respectively).

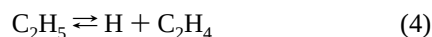
and to the energy transfer parameters.²¹ The low-temperature data of Payne and Stief are relatively close to the high-pressure limit of reaction -1, and therefore, their modeling is most sensitive to the values of $k_{-1}^{\infty}(T)$.

k_{-1}^{∞} is given by^{19,21}

$$k_{-1}^{\infty} = \frac{k_{\text{B}}T}{h} \frac{\kappa(T)Q^{\ddagger}}{Q_{\text{C}_2\text{H}_2}Q_{\text{H}}} \exp\left(-\frac{E_{-1}}{k_{\text{B}}T}\right)$$

where $Q^{\ddagger}$, $Q_{\text{C}_2\text{H}_2}$, and Q_{H} are partition functions of the transition state, acetylene molecule, and H atom, respectively, E_{-1} is the energy barrier height for association, and $\kappa(T)$ is the temperature-dependent tunneling factor (see section III). The ab initio model of the transition state (Tables 2 and 4) has no low vibrational frequencies. As a result, the vibrational partition function of $\text{C}_2\text{H}_3^{\ddagger}$ is essentially equal to 1 at low temperatures (1.07 at 228 K), and therefore, the calculated values of k_{-1}^{∞} are insensitive to variation of vibrational frequencies of the transition state. This leaves only two parameters which affect the high-pressure limit rate constant of reaction -1: the energy barrier E_{-1} and the tunneling correction $\kappa(T)$. In our model the latter is determined by E_1 , E_{-1} , and the barrier width l via formulas I–III and $P(E)$ dependence of Eckart.³⁶ Since l is fixed at the value obtained from the ab initio study and $\kappa(T)$ is relatively insensitive to $\Delta E_{1,-1} = E_1 - E_{-1}$, there is basically only one parameter which determines the calculated value of k_{-1}^{∞} (228 K), the energy barrier E_{-1} .

The fitting procedure is, therefore, reduced to determining three parameters: E_1 , E_{-1} , and some collisional energy transfer characteristic. The latter ($\langle\Delta E\rangle_{\text{down}}$ or $\langle\Delta E\rangle_{\text{all}}$), however, has an unknown temperature dependence which complicates modeling. We repeated the fitting procedure using three different models of the temperature dependence of $\langle\Delta E\rangle_{\text{down}}$. First, $\langle\Delta E\rangle_{\text{down}}$ was taken as independent of temperature; $\langle\Delta E\rangle_{\text{down}} = \text{constant}$. Second, it was assumed to be directly proportional to temperature; $\langle\Delta E\rangle_{\text{down}} = \alpha T$. The latter model is based on the earlier studies of the reaction



The analysis of this system^{45,46} demonstrated a strong dependence of $\langle\Delta E\rangle_{\text{down}}$ on temperature, increasing from 70–100 cm^{-1} at 300 K to 280 cm^{-1} at 1100 K. This dependence was best fit with the directly proportional dependence, $\langle\Delta E\rangle_{\text{down}} = 0.255T \text{ cm}^{-1}$. The analysis of the $i\text{-C}_3\text{H}_7 \rightleftharpoons \text{H} + \text{C}_3\text{H}_6$ system⁴⁷ also supported a strong temperature dependence of $\langle\Delta E\rangle_{\text{down}}$ for He as bath gas.

In the third model of the temperature dependence of $\langle\Delta E\rangle_{\text{down}}$, it was assumed that $\langle\Delta E\rangle_{\text{all}}$ is independent of temperature. This represents as intermediate case of the $\langle\Delta E\rangle_{\text{down}}$ vs T dependence,

one in which $\langle\Delta E\rangle_{\text{down}}$ still increases with T but more slowly than in the case of the directly proportional model.

The above parameters, E_1 , E_{-1} , and the collisional energy transfer parameter ($\langle\Delta E\rangle_{\text{down}}$ or $\langle\Delta E\rangle_{\text{all}}$), were obtained in an optimization procedure. In this procedure, a weighted sum of squares of deviations between the calculated rate constants and the experimental data of Payne and Stief² at 228 K was minimized. Each data point was assigned a weight of $1/k_{-1}$. A constraint requiring agreement between the calculated and experimental (obtained in the current study) k_1 values in the middle of the experimental temperature range (960 K) was imposed. An alternative approach was to fit the Arrhenius $k_1(T, [\text{He}] = 24 \times 10^{16} \text{ atoms cm}^{-3})$ dependence exactly with the constraint that $k_{-1}(T = 228 \text{ K}, P_{\text{He}} = 100 \text{ Torr})$ is reproduced. The resultant values of the fitting parameters obtained with these two approaches are different (Table 5), except for the case of $\langle\Delta E\rangle_{\text{all}} = \text{constant}$.

The best fit to the experimental data was obtained with the third model, $\langle\Delta E\rangle_{\text{all}} = \text{constant}$. The fitted parameters are $E_1 = 158.6 \text{ kJ mol}^{-1}$, $E_{-1} = 16.9 \text{ kJ mol}^{-1}$, and $\langle\Delta E\rangle_{\text{all}} = -74 \text{ cm}^{-1}$. We have chosen this model of reactions 1 and -1 with the above energy parameters and constant value of $\langle\Delta E\rangle_{\text{all}}$ for further use. The rate constants of reactions 1 and -1 calculated using this model are shown in Figures 8 and 9 together with experimental data of this study and of Payne and Stief.² It can be seen from the plots that both high- and low-temperature data are reasonably well reproduced. If model 1 ($\langle\Delta E\rangle_{\text{down}} = \text{constant}$) or model 2 ($\langle\Delta E\rangle_{\text{down}} = \alpha T$) is used, either the temperature dependence of k_1 could be reproduced or the shape of falloff in k_{-1} at 228 K. Although one can reach approximate agreement between the experimental and calculated numbers in the middle of corresponding temperature (for k_1) and pressure (for k_{-1}) intervals, both shapes of these dependencies could not be reproduced together. For illustration purposes, Figure 9 shows (dashed line) the shape of k_{-1} vs pressure dependence obtained with model 2 ($\langle\Delta E\rangle_{\text{down}} = \alpha T$) if the parameters are fitted to reproduce the k_1 vs T dependence. The fitting parameters obtained with all three models of temperature dependence of collisional energy transfer parameters are listed in Table 5.

An attempt was made to fit the sets of data shown in Figures 8 and 9 without accounting for tunneling. In this case, with all three models of the $\langle\Delta E\rangle_{\text{down}}$ vs T dependence, either the temperature dependence of $k_1([\text{He}] = 24 \times 10^{16} \text{ atoms cm}^{-3})$ or the shape of falloff in k_{-1} at 228 K could be reproduced. The shapes of both of these dependencies could not be fitted together. The agreement between the experimental and calculated rate constants was better if a fourth model of $\langle\Delta E\rangle_{\text{down}}$ vs T dependence is used; $\langle\Delta E\rangle_{\text{down}} = \alpha T^{-1/2}$. With this model, both shapes of $k_1([\text{He}] = 24 \times 10^{16} \text{ atoms cm}^{-3})$ vs T and $k_{-1}(228 \text{ K})$ vs $[\text{He}]$ dependencies could be approximately fitted

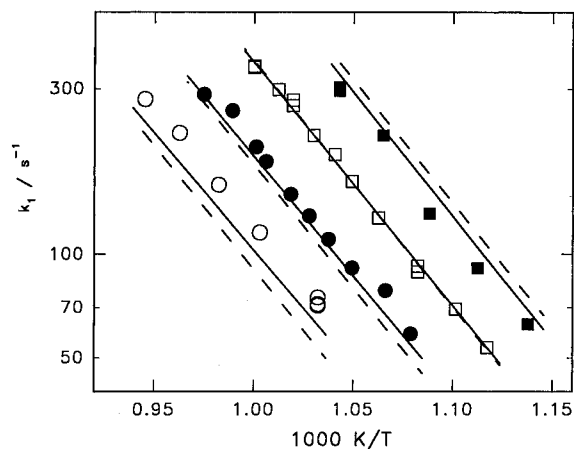


Figure 8. Plot of C_2H_3 unimolecular decomposition rate constants (k_1 vs $1000/T$) for different He densities (atom cm^{-3}). Symbols are the same as in Figure 1. Solid lines represent the results of master equation simulation using our best model with $\langle\Delta E\rangle_{all} = -74$ cm^{-1} . Dashed lines represent a proportional dependence of k_1 on $[He]$ (low-pressure-limit behavior).

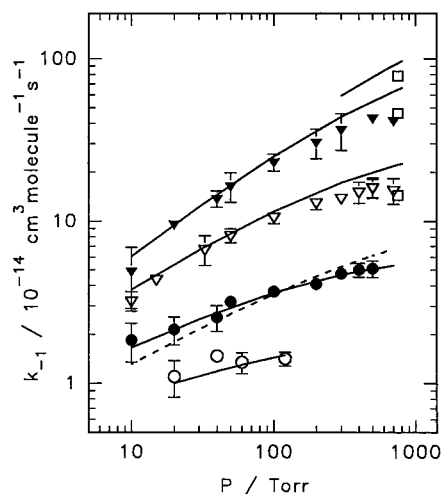


Figure 9. Plot of experimental (symbols, refs 2 and 7) and calculated (lines) rate constants of reaction -1 at different temperatures. Data of Payne and Stief:² open circles, 193 K; closed circles, 228 K; open triangles, 298 K; closed triangles, 400 K. Open squares: data of Ellul et al.⁷ at 298 K (lowest value), 400 K (middle value), and 473 K (highest value). Solid line - master equation fit obtained for the temperatures of experiments using our best model with $\langle\Delta E\rangle_{all} = -74$ cm^{-1} . Dashed line - the shape of k_{-1} vs pressure dependence obtained with model 2 ($\langle\Delta E\rangle_{down} = \alpha T$) if the parameters are fitted to reproduce the k_1 vs T dependence (section IV.3).

with the same parameters. However, this resulted in a significant disagreement at other temperatures of the experiments of Payne and Stief²—193, 298, and 400 K. The calculated rate constants were higher than the experimental ones, on average, by factors of 2 at 298 K and 3 at 400 K. For all four models of $\langle\Delta E\rangle_{down}$ vs T dependence, reproducing the shape of falloff in k_{-1} (228 K) required unrealistically high values of $\langle\Delta E\rangle_{down}$ (228 K)—from 800 to 1400 cm^{-1} —and, in all cases, resulted in similar gross disagreement with experiment at 193, 298, and 400 K.

Our experimental results on $k_1(T, [He])$ are not exactly in the low-pressure limit. Rate constants at $[He] = 24 \times 10^{16}$ atoms cm^{-3} can be represented by an Arrhenius dependence

$$k_1(T, [He] = 24 \times 10^{16}) = 4.79 \times 10^9 \exp(-16400 \text{ K}/T) \text{ s}^{-1} \quad (\text{IV})$$

The dashed lines in Figure 8 show the k_1 vs T dependencies for

the three other densities of He used in our experiments obtained from formula IV and assuming the directly proportional dependence of k_1 on $[He]$. Deviation from the low-pressure-limit dependence can be seen from comparison of these lines with the experimental points. This deviation can be partially explained by the effects of tunneling (see section III). Our best model of reaction 1 (Figure 8, solid lines) partially describes this departure from the low-pressure-limit behavior. This model still yields a somewhat steeper k_1 vs $[He]$ dependence than the experimental one, but not as steep a dependence as in case of no tunneling.

As can be seen on the plot in Figure 9, the data of Payne and Stief on $k_{-1}(T, [He])$ are reasonably well reproduced in the temperature interval 193–400 K by our best model. Although the calculated rate constants are higher than the experimental values at the highest pressures used (by 44% at 298 K and 51% at 400 K), the agreement is good at lower pressures. It should be noted that the parameters of this model were fitted to reproduce the data of these authors at only one temperature, 228 K. The good quality of fit at other temperatures indicates the appropriateness of the model and, in particular, of the set of frequencies (obtained from ab initio calculations) assigned to the transition state. Rate constants at 400 K are more sensitive to these frequencies than those at 228 K. (The $C_2H_3^\ddagger$ vibrational partition function equals 1.5 at 400 K.) Since no experimental information is available on the high-pressure-limit rate constants of reactions 1 and -1 (except for the data of Rao and Skinner¹⁰ which have high uncertainty factors), it is impossible to adjust these frequencies further. We, therefore, accept these ab initio frequencies (Table 4) for further use.

Our best model of reactions 1 and -1 results in the following temperature dependencies of the high-pressure-limit rate constants of reactions 1 and -1 :

$$k_1^\infty = 3.86 \times 10^8 T^{1.62} \exp(-18650 \text{ K}/T) \text{ s}^{-1} \quad (\text{V})$$

$$k_{-1}^\infty = 6.05 \times 10^{-14} T^{1.09} \exp(-1328 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad (\text{VI})$$

at $200 < T < 3000$ K. Expression V does not represent the $k_1^\infty(T)$ dependence very well—the average deviation is 11% with maximum deviation 17%. One can therefore choose to use different rate expressions at lower and higher temperatures

$$k_1^\infty = 7.98 \times 10^5 T^{2.49} \exp(-18316 \text{ K}/T) \text{ s}^{-1} \quad (200 \leq T \leq 1000 \text{ K}) \quad (\text{VII})$$

$$k_1^\infty = 1.13 \times 10^{11} T^{0.92} \exp(-19373 \text{ K}/T) \text{ s}^{-1} \quad (500 \leq T \leq 3000 \text{ K}) \quad (\text{VIII})$$

which give a better representation of $k_1^\infty(T)$ with an average deviation of 3% and maximum deviation of 6%.

Collisional energy transfer parameters for Ar and N_2 (the other bath gases used in our experiments) were obtained from fitting these data with our best model of reaction 1. The numbers obtained for the average energy transferred per collision in both upward and downward transitions are $\langle\Delta E\rangle_{all}(\text{Ar}) = -115$ cm^{-1} and $\langle\Delta E\rangle_{all}(\text{N}_2) = -117$ cm^{-1} . Fitted curves are shown in Figure 2 together with the experimental data.

IV.4. Analysis of Other Studies of Reactions 1 and -1 . Figure 9 also shows k_{-1} values obtained from the Arrhenius dependence reported by Ellul et al.⁷ (these authors did not report the individual $k_{-1}(T)$ values) at 298, 400, and 473 K (open squares). The experiments were conducted at a pressure of 1 bar in a mixture of He and H_2 . Since the fraction of H_2 is not

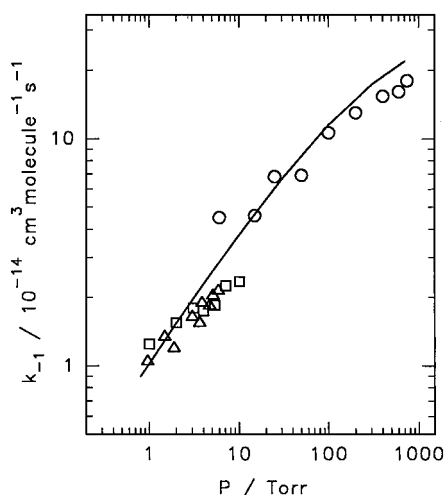
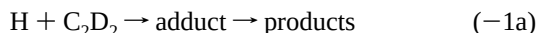


Figure 10. Plot of experimental (symbols, ref 3) and calculated (line) rate constants of reaction -1 at room temperature. Circles represent data obtained in experiments with little or no influence of secondary reactions (stoichiometric factor ≈ 1 , time-resolved photometric experiments). Squares and triangles represent apparent rate constants divided by a stoichiometric factor (≈ 2) to account for influence of secondary reactions (photometric and mass spectrometric experiments, respectively).

reported, it was assumed in the calculations that the bath gas was predominantly helium. Our calculated rate constants (solid lines) lie somewhat higher than the experimental values with the magnitude of the deviation similar to that observed for the data of Payne and Stief.² Calculated constants, however, agree better with the data of Ellul et al. at higher temperatures, a trend opposite to that observed in the case of the data of Payne and Stief.

Figure 10 shows the dependence of k_{-1} vs pressure at room temperature calculated using our best model in comparison with the experimental data of Keil et al.³ Here data obtained under the conditions when contributions of secondary reactions are important are divided by a stoichiometric factor of 2 estimated by these authors. As can be seen from the plot, their experimental data are reasonably well reproduced by the calculations with deviations comparable with the scattering of the experimental data. Our calculated value for the high-pressure-limit rate constant of reaction -1 at 305 K, $k_{-1}^\infty(305 \text{ K}) = 3.7 \times 10^{-13} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$, is in good agreement with the experimental value of Gordon et al.^{27,28} $((3.5 \pm 0.2) \times 10^{-13} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1})$ obtained using two different H-maser techniques.

Rao and Skinner¹⁰ obtained rate constants of reactions 1 and



at $T = 1060\text{--}1900 \text{ K}$ and pressures of $0.3\text{--}3.2 \text{ atm}$ (see section IV.1). While reported values of k_1 were close to the low-pressure limit, those of k_{-1a} were converted by the authors to obtain the high-pressure-limit rate constants k_{-1a}^∞ . We calculated rate constants of reactions 1 and $-1a$ for the conditions of the experiments of Rao and Skinner using our best model of reactions 1 and -1 and the value of $\langle \Delta E \rangle_{\text{all}} = -115 \text{ cm}^{-1}$ for Ar obtained from fitting our experimental data. For the reaction of H atoms with C_2D_2 , rate constant calculations required changes in frequencies, rotational constants, and barrier heights (due to changes in the zero-point vibrational energy). Frequencies and moments of inertia of the $\text{C}_2\text{D}_2\text{H}^\ddagger$ transition state were obtained from our ab initio calculations (frequencies (cm^{-1}): $\text{C}_2\text{D}_2\text{H}$, 528, 676, 835, 954, 1172, 1587, 2148, 2250, 2874; $\text{C}_2\text{D}_2\text{H}^\ddagger$, 448, 607, 614, 632, 755, 1784, 2292, 2636) and those

of C_2D_2 from Suzuki and Overend²⁹ and Lafferty and Thebault.³⁰ Our calculated rate constants of reaction 1 at 0.3 atm of Ar ($k_1 = 2.6 \times 10^3$, 2.8×10^5 , and $1.5 \times 10^6 \text{ s}^{-1}$ at 1000, 1500, and 1900 K, respectively) and reaction $-1a$ in the high-pressure limit ($k_{-1a}^\infty = 2.6 \times 10^{-11}$ and $6.1 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at 1200 and 1900 K, respectively) are, on average, more than an order of magnitude higher than the corresponding values of Rao and Skinner. The results of calculations are outside the error limits reported by these authors, but the differences between calculations and experiment are comparable with the experimental uncertainties of Rao and Skinner which cover more than an order of magnitude in the rate constant values. It should be noted that reaction 1 is closer to the low-pressure limit than to the high-pressure limit under the conditions of experiments of these authors. The ratio k_1/k_1^∞ is 1.2×10^{-3} at 1500 K if our best model is used, and slope of the falloff curve $d(\ln k)/d(\ln p) = 0.92$. Therefore, the calculated rate constants are more sensitive to $\langle \Delta E \rangle_{\text{all}}$ and E_1 values than to the values of k_1^∞ . Better agreement between calculated and experimental values of k_1 can be achieved by using lower values of $\langle \Delta E \rangle_{\text{all}} = -16 \text{ cm}^{-1}$, which brings calculated k_1 within the reported experimental error limits. This significant change in $\langle \Delta E \rangle_{\text{all}}$ (from -115 to -16 cm^{-1}) corresponds to a change in $\langle \Delta E \rangle_{\text{down}}$ from 565 to 146 cm^{-1} at 1500 K. Such a change, however, will result in significant disagreement with our experimental values of k_1 at $[\text{Ar}] = 1.2 \times 10^{17} \text{ atoms cm}^{-3}$. The temperature range of our experimental study in Ar (952–991 K) is very close to the lowest temperature of the experiments of Rao and Skinner (1060 K). It is, therefore, improbable that energy transfer parameters undergo such a significant change between the temperature ranges of our experiments and those of Rao and Skinner. Achieving better agreement between the calculated values of the rate constant of reaction $-1a$ and the data of Rao and Skinner requires an unrealistically tight transition state with very high vibrational frequencies.

Our calculated rate constants are in significant disagreement with the experimental values of Sato et al.^{4–6} This is expected, since we based our model on fitting the results of Payne and Stief which are 2–3 times lower than those of Sato et al. (section IV.1). Disagreement exists not only in the values of high-pressure-limit rate constants, but also in the pressure dependence of k_{-1} . Sugawara, Okazaki, and Sato⁶ observed no pressure dependence of k_{-1} within the pressure range 500–1200 Torr of H_2 at 456 K which led these authors to conclude that reaction -1 is in the high-pressure limit under the condition of their experiments. Our best model of reactions 1 and -1 , however, yields k_{-1}/k_{-1}^∞ values of 0.43 and 0.58 at this temperature and H_2 pressures of 500 and 1200 Torr, respectively. In these calculations, $\langle \Delta E \rangle_{\text{all}} = -117 \text{ cm}^{-1}$ (the same value as determined for N_2 as bath gas) was used. We do not attempt to reproduce these author's data on isotope effects in reaction -1 (obtained at 480–758 Torr) for the following reasons: (1) Different isotopic forms of this reaction will have different k_{-1}/k_{-1}^∞ values because of different energy barriers and, possibly, different values of $\langle \Delta E \rangle_{\text{all}}$. Substitution of hydrogen with deuterium in C_2H_3 will result in a significant change of most vibrational frequencies and, possibly, will affect such properties of the collisions with bath gas as the average energy transferred per collision. (2) While differences in the energy barriers (due to the changes in the zero-point vibrational energy (ZPVE)) can be calculated, such calculations require exact knowledge of the vibrational frequencies of stable species and transition states. A feasible error of 1.2 kJ mol^{-1} (100 cm^{-1}) in the isotopic change of E_{-1} due to the uncertainties in ZPVE will result in a change of $k_{-1}^\infty(298 \text{ K})$ by 38%, which is the

maximum size of the isotope effects measured by Sato et al. at this temperature. (3) The reactions $H + C_2D_2$ and $D + C_2H_2$ are not those of simple recombination but can also have contributions from chemically activated decomposition (to $D + C_2HD$ and $H + C_2HD$) which makes modeling of falloff more difficult. (4) Isotope effects measured by Sugawara et al. are comparable with the difference in rate constant values between the measurements of Sato et al.^{4–6} and those of Payne and Stief.² (The latter are confirmed by other workers.^{3,7}) Factors 1–3, as well as the large scatter in experimental data and the uncertainty in the stoichiometric corrections to the experimental rate constants, prevent us from an analysis of isotope effects studied by Keil et al.³ at low pressures at room temperature.

IV.5. Thermochemistry of Reactions 1 and –1. Kinetic information on reactions 1 and –1 can be used to obtain the enthalpy of reaction and, through this, the enthalpy of formation of the vinyl radical. The recommended values of barriers for these reactions, $E_1 = 158.6 \text{ kJ mol}^{-1}$ and $E_{-1} = 16.9 \text{ kJ mol}^{-1}$, give the energy difference between reactants and products of reactions 1 and –1: $\Delta E_{1,-1} = 141.7 \text{ kJ mol}^{-1}$. Using the models of C_2H_3 and C_2H_2 described above, one can obtain for the enthalpy of reactions 1 and –1 $\Delta H^\circ_{298} = 147.3 \text{ kJ mol}^{-1}$. With $\Delta H^\circ_{298}(C_2H_2) = 228.2 \pm 0.7 \text{ kJ mol}^{-1}$ (ref 48) we obtain $\Delta H^\circ_{298}(C_2H_3) = 298.9 \text{ kJ mol}^{-1}$.

The uncertainty in this value is closely related to all other uncertainties of the model of reactions 1 and –1. We investigate here first the uncertainties of the fitting of data within the assumptions of the model. Second, we estimate the uncertainties associated with the treatment of tunneling and the unknown temperature dependence of $\langle \Delta E \rangle_{\text{down}}$ and $\langle \Delta E \rangle_{\text{all}}$.

The most well determined parameter of the model of reactions 1 and –1 is the barrier for addition of H atoms to acetylene, E_{-1} . This is due to the fact that rate constants of reaction –1 calculated under the conditions of the experiments of Payne and Stief at $T = 228 \text{ K}$ are extremely sensitive to the value of $k_{-1}^\infty(228 \text{ K})$. Within our model, the latter is determined by E_{-1} only. The falloff shape, however, also depends on E_1 and $\langle \Delta E \rangle_{\text{all}}$. The sum of squares of deviations between the experimental data at this temperature and the calculated rate constants was determined as a function of E_{-1} . (E_{-1} was fixed at some values, and the two other parameters were optimized.) From the curvature of this dependence the uncertainty of E_{-1} was estimated⁴⁹ as $\pm 0.2 \text{ kJ mol}^{-1}$ (2σ). If E_{-1} is changed by plus or minus this value, the corresponding changes in E_1 and $\langle \Delta E \rangle_{\text{all}}$ required to reproduce our data on k_1 are $+3.5/-1.8 \text{ kJ mol}^{-1}$ and $-36/+13 \text{ cm}^{-1}$, respectively.

The uncertainties originating from the treatment of tunneling were estimated by repeating the fitting process with the width parameter of reaction barrier l changed by $\pm 0.23 \text{ amu}^{1/2} \text{ \AA}$, the difference between l values obtained from ab initio calculations at PMP4 and UMP2 levels. The corresponding changes in E_1 , E_{-1} , and $\langle \Delta E \rangle_{\text{all}}$ are $+3.2/-4.3 \text{ kJ mol}^{-1}$, $-0.4/+0.5 \text{ kJ mol}^{-1}$, and $-41/+33 \text{ cm}^{-1}$. Combining the uncertainties resulting from these two reasons, we obtain for the heat of formation of vinyl radical and average energy transferred per collision:

$$\Delta H^\circ_{f, 298} = 298.9^{+6.4}_{-6.9} \text{ kJ mol}^{-1}$$

$$\langle \Delta E \rangle_{\text{all}} = -(74^{+77}_{-46}) \text{ cm}^{-1}$$

It should be stressed here that these uncertainties are estimated within the framework of the current model, i.e., using the one-dimensional Eckart function to describe tunneling and assuming that $\langle \Delta E \rangle_{\text{all}}$ is independent of temperature. If other temperature dependencies of $\langle \Delta E \rangle_{\text{all}}$ or $\langle \Delta E \rangle_{\text{down}}$ are used, changes in barrier

height values will be necessary (Table 5). Therefore, if the temperature dependence of energy transfer parameters is presumed unknown, that will introduce an additional uncertainty into the value of the heat of formation of vinyl radical: $+7.2/-7.7 \text{ kJ mol}^{-1}$. In this case, combining the above components of uncertainties, we obtain

$$\Delta H^\circ_{f, 298} = 298.9^{+13.6}_{-14.6} \text{ kJ mol}^{-1}$$

V. Discussion

The current study presents the first direct experimental measurement of the rate constants of unimolecular decomposition of vinyl radical. Master equation/RRKM modeling of these experimental results and of data on the reverse reaction, the addition of H atom to acetylene, available in the literature resulted in the creation of a model of reactions 1 and –1 that describes its kinetics at both low (193–400 K) and high (879–1058 K) temperatures and in different parts of the falloff. It is shown that tunneling plays an extremely important role in the mechanism of the reaction under study. The influence of tunneling is manifested by significantly larger changes in the low-pressure-limit rate constants and in the shape of falloff than in the values of the high-pressure-limit rate constants.

The best agreement between experimental and calculated rate constants was obtained when the average energy transferred per collision in both upward and downward directions, $\langle \Delta E \rangle_{\text{all}}$, is assumed to be independent of temperature for He as bath gas. Such a $\langle \Delta E \rangle_{\text{all}} = \text{constant}$ form of the temperature dependence of collisional parameters has been advocated by several groups. A temperature-independent $\langle \Delta E \rangle_{\text{all}}$ is consistent with the results of experimental and trajectory studies of collisional energy transfer in large polyatomic molecules.^{50,51} The recent results of Kiefer et al.⁵² on the thermal decomposition of CF_3Cl in Kr at temperatures 1521–3059 K were best fit with the constant value of $\langle \Delta E \rangle_{\text{all}} = -270 \text{ cm}^{-1}$. Dashevskaya et al.⁵³ predict $\langle \Delta E \rangle_{\text{all}} = \text{constant}$ dependence for He as a bath gas in the framework of the sequential direct encounter model.

The overall agreement between our best model of reactions 1 and –1 and the experimental results of Payne and Stief,² Keil et al.,³ Ellul et al.,⁷ and Gordon et al.^{27,28} is good. Although the reaction model properties were selected to reproduce the data of Payne and Stief at one temperature only (228 K), the agreement with experimental data at other temperatures is satisfactory (Figures 9 and 10). While at the highest experimental pressures (close to 1 atm) the calculated rate constants are somewhat higher than the experimental ones, the agreement is better at lower pressures. At the same time, the calculated $k_{-1}^\infty(305 \text{ K})$ coincides with the experimental value of Gordon et al.^{27,28}

Earlier attempts^{3,8} to reproduce the experimental temperature and pressure dependencies of the rate constant of reaction –1 yielded rate constants values at least an order of magnitude higher than experimental ones. It was concluded that no reasonable model of an activation complex could account for the experimental preexponential factor. One of the main reasons for this failure of theory to agree with experiment is that Keil et al.³ and Harding et al.⁸ attempted to reproduce the k_{-1}^∞ Arrhenius dependence reported by Payne and Stief.² This dependence ($9.2 \times 10^{-12} \exp(-1213 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$) represents values of k_{-1} obtained at the highest pressure at each temperature of the experiments of Payne and Stief. As can be seen from Figure 9, the data of these authors are in the falloff, and rate constants obtained at higher temperatures are further from the high-pressure limit than are those obtained at low temperatures. Therefore, the reported Arrhenius dependence

has a lower activation energy and preexponential factor than does the $k_{-1}^\infty(T)$ dependence. Our best model of reactions 1 and -1 yields the $k_{-1}^\infty(T)$ dependence described by expression VI or, in the temperature range of the experiments of Payne and Stief, $k_{-1}^\infty(T) = 7.84 \times 10^{-11} \exp(-1596 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$.

Another reason for the earlier disagreement between theory^{3,8} and experiment is that prior theoretical studies did not account for the significant influence of tunneling on the shape of falloff, in particular, the widening of falloff and its extension toward lower pressures (section III). It is shown (section IV.3) that experimental data on k_{-1} cannot be reproduced without accounting for tunneling. Any attempt to fit the data at one temperature results in gross disagreement (factor of 2–3) at other experimental temperatures, i.e., the experimental low-temperature (193–400 K) preexponential factor and activation energy do not match those obtained in calculations.

Kinetic information on reactions 1 and -1 is used to calculate the heat of formation of vinyl radical. The value obtained in our calculations is $\Delta H_f^\circ_{298}(\text{C}_2\text{H}_3) = 298.9 \text{ kJ mol}^{-1}$ with error limits depending on whether the average energy transferred per collision $\langle \Delta E \rangle_{\text{all}}$ is assumed to be independent of temperature ($+6.4$ to -6.9 kJ mol^{-1}) or have an unknown temperature dependence ($+13.6$ to $-14.6 \text{ kJ mol}^{-1}$). Significantly divergent values of $\Delta H_f^\circ_{298}(\text{C}_2\text{H}_3)$ can be obtained from the literature (see the review in ref 31). Analysis of the kinetic data^{54,55} on the reaction $\text{C}_2\text{H}_3 + \text{HCl} \rightarrow \text{C}_2\text{H}_4 + \text{Cl}$ yields $\Delta H_f^\circ_{298}(\text{C}_2\text{H}_3) = 279.9 \pm 1.3 \text{ kJ mol}^{-1}$, while a method based on gas-phase acidity results³¹ in $\Delta H_f^\circ_{298}(\text{C}_2\text{H}_3) = 300.0 \pm 3.3 \text{ kJ mol}^{-1}$. Our value of $\Delta H_f^\circ_{298}(\text{C}_2\text{H}_3)$ coincides with the latter value determined by Ervin et al.³¹ within the experimental error limits. A recent study of the formation of vinyl radical in reaction $\text{Cl} + \text{C}_2\text{H}_4$ by Kaiser and Wallington⁵⁶ resulted in the value of $295.4 \pm 1.7 \text{ kJ mol}^{-1}$, which is in good agreement with our result.

The theoretical treatment of the falloff shape (section III) indicates a significantly larger influence of tunneling on the rate constants in the falloff and low-pressure limit than on those in the high-pressure limit. These effects are important even at high temperatures where high-pressure-limit tunneling contributions are negligible. Unfortunately, the current treatment of tunneling as a one-dimensional motion with an Eckart potential is only a crude approximation. The demonstrated importance of tunneling indicates a necessity to develop better methods of accounting for it in the modeling of falloff.

VI. Parametrization of Rate Constants

We present here a parametrization of k_1 in helium and nitrogen which provides rate constant values throughout the range of temperatures and pressures 200–2000 K and $1\text{--}10^4$ Torr. The modified Lindemann–Hinshelwood expression introduced by Gilbert et al.⁵⁷ was used:

$$\frac{k_1}{k_1^\infty} = \frac{k_1^0[M]/k_1^\infty}{1 + k_1^0[M]/k_1^\infty} F(F_{\text{cent}}, [M]) \quad (\text{IX})$$

Values of $k_1([M], T)$ at different temperatures and pressures were calculated using the master equation/RRKM approach assuming $\langle \Delta E \rangle_{\text{all}}$ independent of temperature for both bath gases. The values of $\langle \Delta E \rangle_{\text{all}}$ obtained from the modeling of experimental data (-74 cm^{-1} in He and -117 cm^{-1} in N_2) were used. The calculated temperature dependencies of high- and low-pressure-limit rate constants are

$$k_1^\infty = 3.86 \times 10^8 T^{1.62} \exp(-18650 \text{ K}/T) \text{ s}^{-1} \quad (\text{V})$$

(see also expressions VI and VIII, section IV.3) and

$$k_1^0 = 6.58 \times 10^3 T^{-3.49} \exp(-18070 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad (\text{in He})$$

$$k_1^0 = 4.26 \times 10^3 T^{-3.40} \exp(-18021 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad (\text{in N}_2)$$

The values of $k_1([M], T)$ were fitted with the modified Lindemann–Hinshelwood expression IX. It was found that values of $\log(F_{\text{cent}})$ increase linearly with $\log(T)$:

$$F_{\text{cent}} = 7.94 \times 10^{-4} T^{0.78} \quad (\text{in He})$$

$$F_{\text{cent}} = 7.37 \times 10^{-4} T^{0.80} \quad (\text{in N}_2)$$

One should note here that formula IX does not give a very good representation of falloff in reaction 1 at temperatures below 1000 K. The average deviation of the parametrized values of k_1 from the master equation rate constants over the large range of conditions used is 16%, with maximum deviation of 41%.

Recently, Bernshtein and Oref⁵⁸ demonstrated that the steady state approximation cannot be used for calculation of unimolecular reaction rates by the master equation solution at high temperatures and low pressures where relaxation to the steady state population occurs on time scales comparable to the reaction time. The examples given by these authors indicate that reasonable agreement between the exact values of rate constants and those obtained under the steady state assumption is reached after several hundred collisions with the bath gas. The calculations presented here are limited to conditions where the ratio of collision frequency ω to the rate constant is higher than 400. This requirement and the maximum pressure of 10^4 Torr determine the upper temperature limit (2120 K) of the current parametrization. For the low-pressure-limit rate constant expressions, however, the upper temperature limit determined by this requirement is lower—1960 K for He and 1830 K for N_2 . At any temperature between 1830 and 2120 K the pressure range of applicability of the above expressions can be obtained from comparing the parametrized rate constants with the collision frequency.

Rate constants of addition of H atoms to C_2H_2 , reaction -1 , can be found from the current parametrization of k_1 and the equilibrium constant K_{eq} via^{21,22}

$$k_{-1} = k_1 K_{\text{eq}} f_{\text{ne}}$$

The correction factor f_{ne} was found to be nearly equal to 1 for all the conditions of the current parameterization where the above criterion of applicability of the steady state master equation is satisfied. The lowest value of $f_{\text{ne}} = 0.81$ was obtained at 1 Torr at 2000 K where $\omega/k_1 = 400$.

VII. Summary

Rate constants of the unimolecular decomposition of vinyl radical (reaction 1) in helium, argon, and nitrogen have been obtained near the low-pressure limit. The transition state of reactions 1 and -1 and the shape of the potential energy profile along the reaction path were studied using ab initio methods. In a separate theoretical section, substantial effects of tunneling on the shape of falloff and the values of the low-pressure-limit rate constants were investigated.

In the data analysis, the experimental rate constants of reaction 1 (obtained in this study) and reaction -1 (reported by other workers^{2,3,7,27,28}) were reproduced by a master equation model-

ing. The geometry and vibrational frequencies of the transition state were taken from the results of the ab initio study. Values of $k(E)$ were calculated by the RRKM method. The effects of tunneling were included using the analytical expression of Eckart for the transition probability $P(E)$ with one of the parameters (barrier width l) obtained from the ab initio calculations and two other parameters (barriers for the direct and reverse reactions E_1 and E_{-1}) obtained from fitting of the experimental data.

With other parameters of the transition state fixed at the values obtained from the ab initio calculations, k_1 and k_{-1} are determined by the following three parameters only: E_1 , E_{-1} , and $\langle \Delta E \rangle_{\text{down}}$. Since the latter has an unknown temperature dependence, three different functional forms of this dependence were used: $\langle \Delta E \rangle_{\text{down}} = \text{constant}$ (model 1), $\langle \Delta E \rangle_{\text{down}} = \alpha T$ (model 2), and $\langle \Delta E \rangle_{\text{all}} = \text{constant}$ (model 3). Within each of these models, values of $\langle \Delta E \rangle_{\text{down}}$ (or $\langle \Delta E \rangle_{\text{all}}$), E_1 , and E_{-1} were obtained from the fitting of the experimental data. In the fitting procedure, attention was focused on reproducing the experimental data of Payne and Stief² on k_{-1} at 228 K and those on k_1 obtained in the current study at $[He] = 24 \times 10^{16}$ atoms cm^{-3} . The best agreement was achieved with the third model of the $\langle \Delta E \rangle_{\text{down}}$ vs T dependence, i.e., $\langle \Delta E \rangle_{\text{all}} = \text{constant}$. The master equation modeling of the experimental data resulted in the value of $\Delta H_f^\circ_{298}(C_2H_3)$ which is in good agreement with the latest literature data. Finally, parameterized expressions are presented for obtaining k_1 in helium and nitrogen over a wide range of pressures and temperatures.

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